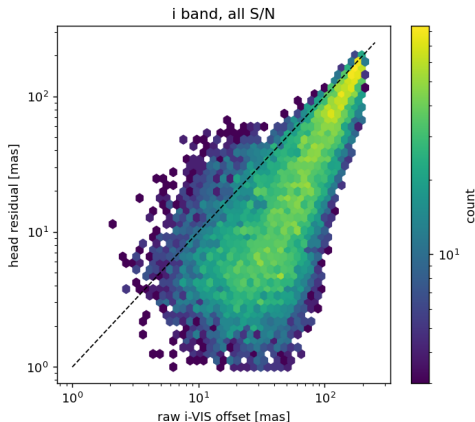
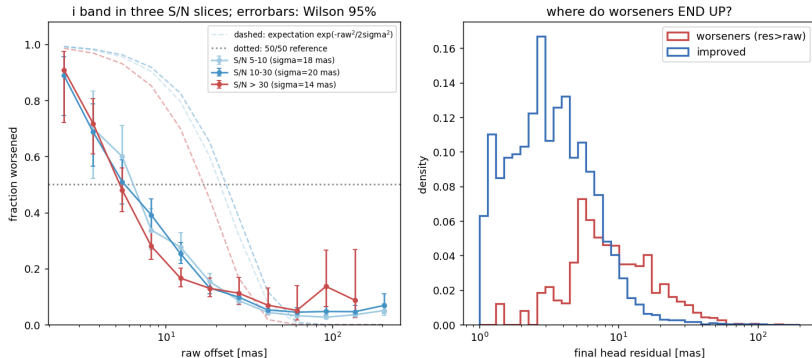


Why Astrometry Is Degraded for a Subset of Objects: Diagnosis and Cure in JAISP

July 2026, updated August 2026 (internal, ECDFS Q1)



- The head (**v10 production**) improves the population (raw ~ 48 mas $\rightarrow$ 13.4 mas median, all bands; panel: i band), but 9% of all measurements end up **farther** from VIS than they started.
- Two candidate explanations: (a) those sources were lucky before, and the small raw offset was chance; (b) the model fails on those sources specifically.



v10 production head

- Left: the same pattern in every S/N slice, each observed curve below its own re-measurement expectation (dashed, $P(\text{worse}) = e^{-\text{raw}^2/2\sigma^2}$); pooled over bands, shared photons preserve lucky agreement more often than chance predicts.
- Only the **bright** slice grows a tail at large raw offsets, where chance predicts zero. Right: worseners sit right of the improved because the selection forces it (res > raw truncates the floor distribution from below); the truncated-floor prediction is 21.6 mas median versus 17.8 observed. Only the excess beyond ~ 40 mas escapes any chance model

For one source, per band b : fit the classical centroid, form $\text{raw}_b = \text{pos}_b - \text{pos}_{\text{VIS}}$; the head (v10) predicts a correction c_b from one shared representation; $\text{resid}_b = \text{raw}_b - c_b$. “Strongly worsened in b ” = $|\text{resid}_b| > 1.5 |\text{raw}_b|$ and worse by > 10 mas. Rate: 5.6% of bright measurements.

- **Independent between bands:** the centroid-fit noise of each band image (different exposures, different photons). If this alone drove worsening, hits on the same source would be independent coin flips: $P(3 \text{ hits}) \approx 0.056^3 \times 10 \text{ triples} \times 3185 \text{ sources} \approx 5$.
- **Shared for one source:** (1) the VIS reference position, subtracted from every band; (2) the local WCS terms (Rubin bands share Rubin’s solution, NISP its own); (3) the source itself (proper motion, colour gradients); (4) the head’s shared representation, feeding all nine corrections.

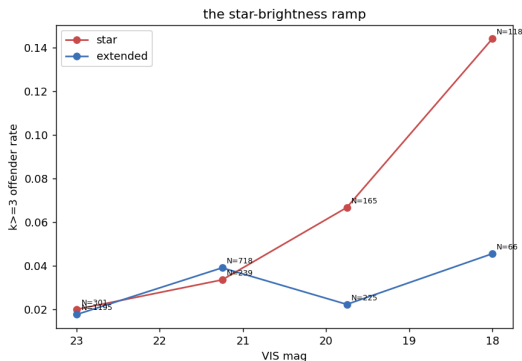
strongly worsened in $\geq k$ Rubin bands	observed	chance (permutation null)
$k \geq 2$	181	68 ± 7
$k \geq 3$	113	5 ± 2
$k \geq 4$	61	0.2
$k \geq 5$	23	0.0

The null shuffles flags within (band, raw offset, S/N) classes, preserving all rates and the VIS-induced raw correlations (hence chance $k \geq 2$ is 68, not 0). Robust to the cuts (S/N>10: 200 vs 8; all nine bands: 153 vs 12) and to the flag definition: the excess exists at every threshold. Tightening just cleans the list of chance interlopers (fiducial: 113 flagged, only ~ 6 innocent), making it clean enough to study source by source.

Stage 1 established a source-attached, band-shared cause, not yet necessarily the head. Four things are shared; each gets a measurement:

- **(1) The VIS reference?** A bad pos_{VIS} would worsen all bands at once, a plain measurement error. Gaia says no: for the offender stars the VIS labels sit **7.0 mas** from the PM-propagated Gaia position, better than the survey average.
- **(2) The WCS?** A solution error is smooth in sky position: it cannot single out one star and spare its neighbours. Offenders show **no sky clustering** (nearest-offender distance $62''$ vs $66'' \pm 5''$ for random draws), and the failure **persists in NISP** (59% of offenders worsened in all NISP bands), whose astrometric solution is independent of Rubin's.
- **(3) The source itself?** Proper motion correlates with nothing ($\rho = 0.01$ against residuals); high-PM stars are actually the head's success cases (it bridges the epochs).
- **(4) The head's representation** carries positive fingerprints, not just survival by elimination: corrections **larger than the disagreement they correct**, directions coherent across all nine bands and **both instruments** (one wrong belief applied everywhere), and Gaia places the head far from truth where every classical measurement of the same stars is at the floor (numbers on the next slides).

The two-stage verdict: repetition proves the cause is source-attached and shared; elimination plus fingerprints prove it is the head. The 113 are 0.3% of the catalog, identifiable one by one. (All measurements on this slide: v10 production head.)



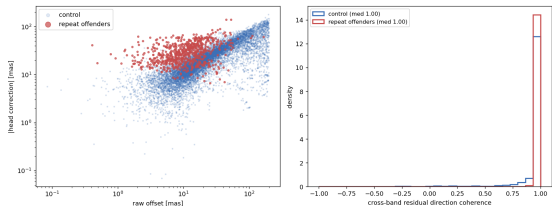
Offender rate (**v10 production head**) for stars ramps 2% $\rightarrow$ **14%** from VIS 24 to VIS 17; extended sources stay flat.

Population: the 3,185 sources with ≥ 2 bright (S/N>20) Rubin bands, the only ones the k statistic can score, out of 37,796 in the catalog; N labels are the per-bin denominators of that eligible population, and rate $\times$ N summed over bins and classes recovers the 113.

Further mechanisms tested and **ruled out** (beyond the WCS and proper motion of the previous slide):

- tile-edge context (paired same-source test: ± 0.1 mas)
- companions within $2''$ (13% vs 20% control)
- VIS saturation (onset ≈ 17 , masked not clipped; offender cores pristine)

Against PM-propagated Gaia DR3 (epoch 2024.83), per star:



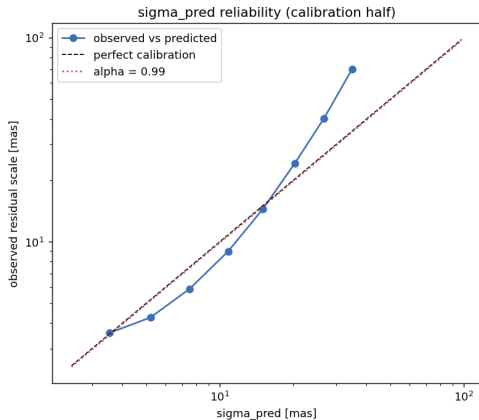
	VIS label	head (v10)
offender stars ($k \geq 3$)	7.0 mas	22.9 mas
all Gaia stars	9.5 mas	12.2 mas

Left: offenders receive corrections **larger than the raw disagreement** ($|c| \sim 28$ mas vs raw ~ 11 mas), coherently in every band: the head asserts a center the classical centroids contradict.

Reading (as of July): a brightness-dependent divergence between the Gaussian-fit convention and the head's feature-encoded center. *The actual mechanism was found in August — slides ahead.*

JULY 2026: INTERIM MITIGATION — superseded by the v11 foundation fix (slides 12–17)

Two estimators of the same VIS-convention position: classical (error scale σ_{cl} from S/N and class) and head-corrected (the head's own σ_{pred}).

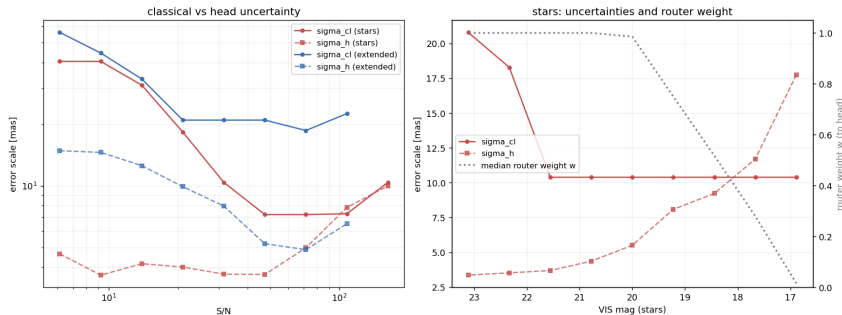


$$\text{combined} = (1 - w) \text{raw} + w \text{resid},$$

$$w = \frac{\sigma_{\text{cl}}^2 - \rho \sigma_{\text{cl}} \sigma_h}{\sigma_{\text{cl}}^2 + \sigma_h^2 - 2\rho \sigma_{\text{cl}} \sigma_h}$$

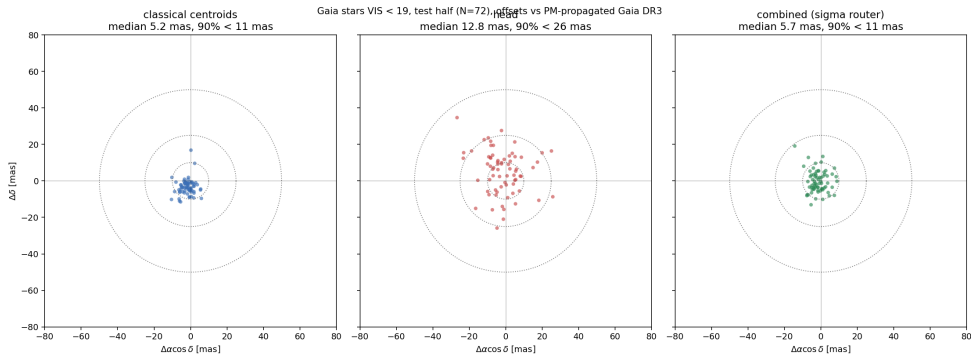
- σ_{pred} is nearly perfectly calibrated in bulk ($\alpha = 0.99$); the over-confidence lives only in the offender tail.
- The correlation matters: head and classical errors share photons ($\rho \approx 0.8$ extended, 0.5 stars); naive inverse variance costs 2 mas at the faint end, the correlation-aware weight costs nothing.

JULY 2026: INTERIM MITIGATION — superseded by the v11 foundation fix (slides 12–17)

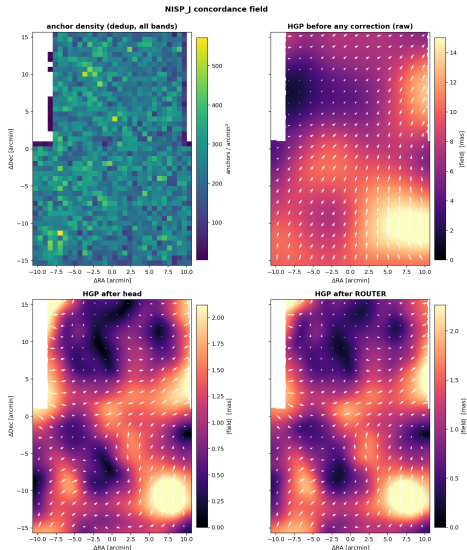


- Extended sources: classical centroiding is morphology limited ($\sigma_{cl} \gtrsim 20 \text{ mas}$ at every S/N); the curves **never cross**, so galaxies are pure head.
- Stars: crossing at S/N ≈ 108 (VIS ≈ 18.2); the weight (gray) slides $1 \rightarrow 0$ across VIS $20 \rightarrow 17$. The magnitude gate is **derived from the uncertainties**, not imposed.
- σ_h **rises** for bright stars: the uncertainty channel partially knows the head struggles there; the offenders are the tail where that self-awareness fails.

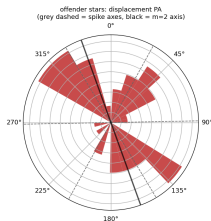
JULY 2026: INTERIM MITIGATION — superseded by the v11 foundation fix (slides 12–17)



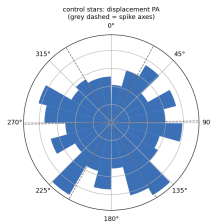
median radial offset [mas]	classical	head	router
all sources	50.6	14.6	14.7
bright stars, vs Gaia truth	5.2	12.8	5.4
faint (S/N < 10)	64.9	18.5	18.5



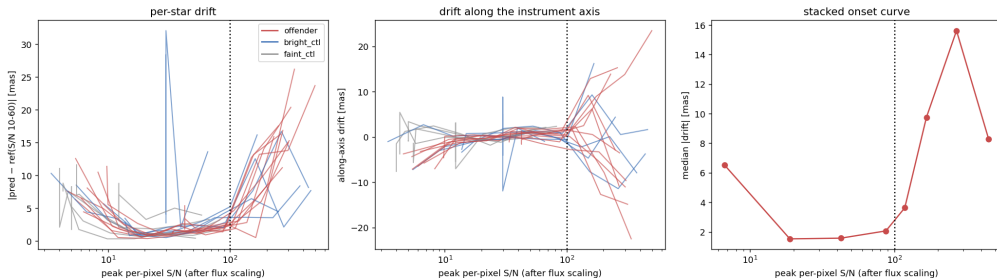
- Refit the field (HGP super-tight, PINN) on router residuals (v10-era archives): median change **0.16 mas**, an order of magnitude below the field amplitude. The robust fit had already absorbed the 113 outliers; Figure 10 and the FITS products stand.
- The ~ 15 mas lobe at lower right is **real**: coherent in 8 of 9 bands relative to VIS, normal anchor coverage, both solvers agree, and it sits within MER tile TILE102043549. Eight rulers agree and one does not: a localized feature of the **VIS solution**.
- By construction no VIS-trained correction can remove it; the released raw-field FITS is precisely the map that carries it.



v10 archive (ECDFS)



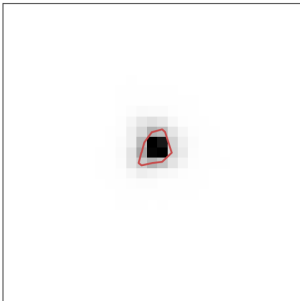
- Offender displacements are **elongated along a single sky axis** (PA $\approx 160^\circ$, both signs, no net direction); controls are isotropic.
- The axis **rotates with the Euclid instrument** between ECDFS and EDF-S (spike phases $28.9^\circ \rightarrow 49.7^\circ$).
- Published VIS geometry (strut angles + four-corner CCD readout) predicts the readout axis **1.9°** from the measured one, with zero tuned parameters — in both fields, one parity.
- Yet a signed flux stack shows **no trailed light** (CTI ruled out), and rotating a star's own pixels rotates its displacement: the cause lives in the images as the network sees them.



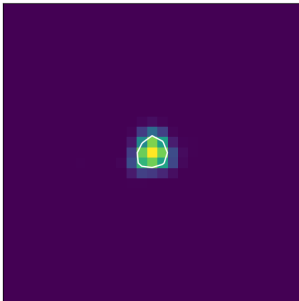
causal interventions on the v10 production model

- Every input enters the foundation as `(image/rms).clamp(-10, 100)` — unchanged since v4. Above per-pixel S/N 100 a star's core is a **featureless plateau**; the reconstruction targets are clamped identically, so **no loss ever saw the missing information**.
- Causal test: scale a real star's flux $\times 50$ — the head places the *same star* to 1–2 mas for peak S/N 10–90, and the displacement **switches on exactly above $S/N \approx 100$** (a threshold, not a trend), reaching the archived 11–26 mas at native flux.
- Rotating the star's local image 180° flips the displacement (median $\cos = +0.91$): each star's error is deterministic, written in its own pixels.

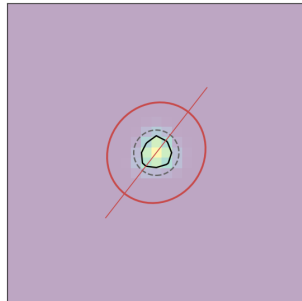
one bright star (VIS=18.9)
red: S/N=100 contour = clamp boundary



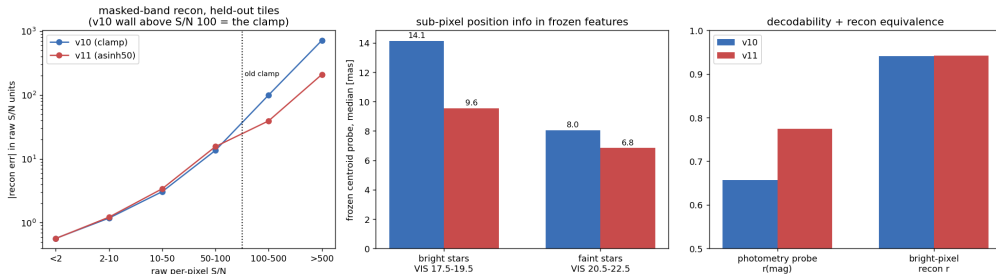
stacked clamp mask, 131 stars
(recentered occupancy fraction)



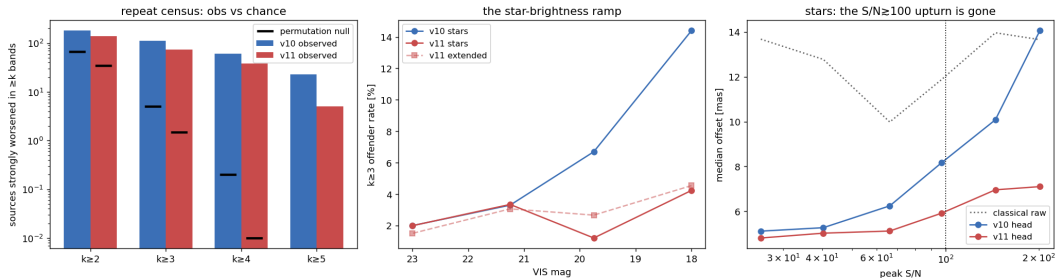
fitted ellipse: ratio 1.085, PA 142°
(red) vs circle (dashed); axis drawn



- The clamp is isotropic, but the surviving information is not: per star the S/N=100 plateau (left, red) is a ragged blob, yet the **stack of 131 masks** (middle) is measurably elliptical — **axis ratio 1.085** (CI [1.05, 1.20]), **major axis PA 142° ± 15°**, consistent with the offender displacement axis (160° ± 9°) and the instrument frame.
- An elongated boundary constrains the centroid weakly along its major axis: errors along one axis, both signs, no net drift — the measured $m=2$ -without- $m=1$ signature; the flux stack shows **no trailed light** (CTI ruled out).
- Under v11 the same elongation is still in the images — with the core restored, the head stops leaning on the boundary and the axis signal vanishes. Per-star sign mechanism (subpixel phase?) untested.



- `clamp(-10,100) → 50*asinh(x/50)` in stems *and* reconstruction targets; foundation retrained from scratch, identical recipe (`jaisp_v11_q1_soft`).
- Gate: reconstruction parity in bulk (bright-pixel r 0.9412 vs 0.9419); the v10 error **wall above S/N 100** is gone (left); frozen centroid probe on bright stars **14.1 → 9.6 mas** (middle); photometry decodability up.
- The position head is retrained **unchanged** — no side-channel, no reweighting. The architecture was never the problem; its inputs were.



- The star-brightness ramp is **flat** (2–4% at all magnitudes, indistinguishable from galaxies; was 2 → 14.4%).
- The $S/N \gtrsim 100$ upturn is **reversed**: the v11 head sits at 5–7 mas where classical raw is 10–14 — it now adds value on bright stars. General Gaia stars: head at label parity (10.8 vs 9.5 mas).
- Faint-end headline untouched (raw 47.9 → 13.3 mas, 91% improved — identical to v10). Under v11 the router no longer improves any regime: the interim mitigation is **retired by the cure**.

Decomposition of the 113 repeat offenders:

~46 clamp-driven bright stars	cured by v11	(the ramp, the upturn, the axis)
~67 source-specific cases	survive both foundations	(the new open question)

The survivors: the *same sources* under two independently trained foundations, same signature (raw ~ 10 mas, coherent ~ 30 mas head displacement; Gaia still against the head, 29.5 vs 9.6 mas for the 8 Gaia stars among them). Not brightness-selected, not resolved blends (12% vs 20% control), not extended.

- Cause is outside the foundation: in the data or labels of these specific sources, or in what both trainings share (label pipeline, tiles). Suspects: **sub-arcsecond pairs below MER's deblending limit**; convention-ambiguous profiles with unstable training labels.
- 0.2% of the catalog; the uncertainty router continues to cover them in the meantime.

- Above the 1:1 line sit two populations: mostly benign regression to the corrected floor, plus real failures concentrated in bright point sources — diagnosed by **repetition across bands**, not by any average metric.
- The mechanism was one line of preprocessing: a hard per-pixel S/N clamp that erased bright-star cores — invisible to every loss curve because the targets were clamped too. **Audit your input transforms; stratify metrics by regime.**
- Correlational forensics narrowed the suspects; **causal interventions** (flux sweep, rotation) convicted. The fix is soft compression + retrain (v11): ramp flat, upturn reversed, faint-end headline untouched, detection deeper.
- The interim uncertainty router did its job and is retired by the cure; 67 source-specific cases remain — the next investigation.
- The paper sentence, updated: the head improves astrometry where classical methods are noise limited, and now **matches or beats them** where they are systematics limited.